

Polarization domain walls in diffractive or dispersive Kerr media

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A novel type of vector dark solitary wave is shown to exist in Kerr materials. These solitary waves consist of localized structures separating adjacent regions (or domains) of different polarization eigenstates of the Kerr medium.

In an accounting for the polarization of the electromagnetic field, propagation of light in Kerr materials is ruled by coupled nonlinear Schrödinger (NLS) equations. These equations were shown to exhibit a much richer dynamics than the single NLS equation.¹ In particular, the idea that the linear and nonlinear polarization coupling processes could sustain bound solitary waves belonging to two polarization modes was first considered by Christodoulides and Joseph² and independently by Tratnik and Sipe.³ These authors showed that bound states of two orthogonally polarized bright solitons can propagate in fibers in the anomalous dispersion regime thanks to the interplay between linear birefringence and cross-phase modulation. Christodoulides⁴ verified that the coupling of bright and dark solitons may also result from this process. We showed in a recent study that, in fact, birefringence is not necessary for such bound soliton states to exist.^{5,6} Finally, considering highly birefringent media with normal dispersion, Kivshar and Turitsyn showed the existence of bound states of gray solitons.⁷ In this Letter we show that, on the basis of a simple analysis of the coupled NLS equations, Kerr media can sustain localized structures separating domains of different nonlinear polarization eigenstates (or eigenpolarizations). In the context of nonlinear spatial phenomena they are transverse structures of a two-dimensional beam in a self-defocusing material. In the temporal domain these structures propagate in the normal dispersion regime in media with fast positive Kerr nonlinearity and one-dimensional geometry (such as single-mode fibers).

In normalized units the propagation of the linear polarization components of a field in Kerr media is described by the coupled NLS equations (see, for example, Refs. 1–3)

$$i \frac{\partial U}{\partial z} + \frac{1}{2} \frac{\partial^2 U}{\partial x^2} - [|U|^2 + (1-B)|V|^2]U - BV^2U^* + \kappa U = 0, \quad (1a)$$

$$i \frac{\partial V}{\partial z} + \frac{1}{2} \frac{\partial^2 V}{\partial x^2} - [|V|^2 + (1-B)|U|^2]V - BU^2V^* - \kappa V = 0, \quad (1b)$$

where $U(x, z)$ and $V(x, z)$ are the envelopes of both polarization components, κ is the normalized bire-

fringence parameter, and $B = \chi_{1221}^{(3)}/\chi_{1111}^{(3)}$, where $\chi_{ijkl}^{(3)}$ is the nonlinear susceptibility tensor of the material. The coordinate x represents either the transverse spatial coordinate of a two-dimensional beam in a self-defocusing medium or the time coordinate in a reference frame traveling at the group velocity of a pulse in the normal dispersion regime. Note that, for the temporal problem, the difference in the group velocities of both polarization modes is neglected (see Ref. 8 for an analysis of this approximation).

We look for bound solitary-wave solutions in which both polarization components have the same propagation constant and are $\pi/2$ out of phase:

$$\begin{aligned} U(x, z) &= u(x)\exp(-i\beta z), \\ V(x, z) &= iv(x)\exp(-i\beta z). \end{aligned} \quad (2)$$

The envelopes u , v and the propagation constant β are real. Substitution of these forms of U and V into Eqs. (1) leads to a set of coupled ordinary differential equations:

$$\frac{1}{2} \ddot{u} + \beta u - u^3 - (1-2B)v^2u + \kappa u = 0, \quad (3a)$$

$$\frac{1}{2} \ddot{v} + \beta v - v^3 - (1-2B)u^2v - \kappa v = 0, \quad (3b)$$

where the overdots denote x derivatives. Equations (3) are the equations of motion in the plane (u, v) of a unit mass in the potential

$$\begin{aligned} V(u, v) &= \beta(u^2 + v^2) - \frac{1}{2}(u^2 + v^2)^2 \\ &\quad + 2Bu^2v^2 + \kappa(u^2 - v^2). \end{aligned} \quad (4)$$

This potential possesses maxima and saddle points that correspond to the uniform (plane-wave or cw) stationary solutions to Eqs. (1). The polarization states of these waves constitute the eigenpolarizations of the birefringent Kerr material. Setting $\ddot{u} = 0$ and $\ddot{v} = 0$ in Eqs. (3) we can easily calculate these points. We find four saddle points on the u and v axes: $(u = \pm\sqrt{\beta + \kappa}, v = 0)$ and $(u = 0, v = \pm\sqrt{\beta - \kappa})$. They represent linear eigenpolarizations

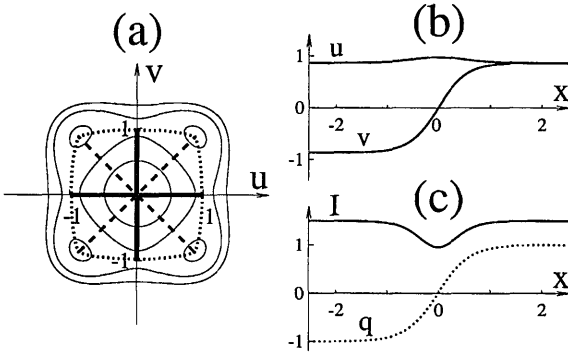


Fig. 1. (a) Contour plot of the potential V in the plane (u, v) for $\beta = 1$. The thick solid lines and dashed lines correspond to the linearly and the circularly polarized NLS dark solitons, respectively. The dotted curves that connect adjacent maxima correspond to the polarization domain walls. (b) The envelopes u and v of the domain walls. (c) Intensity profile I and ellipticity degree q of the domain wall.

parallel to the fast and slow axes of the birefringent Kerr material. The potential V also possesses four maxima:

$$u = \pm \left[\frac{1}{2} \left(\frac{\beta}{1-B} + \frac{\kappa}{B} \right) \right]^{1/2},$$

$$v = \pm \left[\frac{1}{2} \left(\frac{\beta}{1-B} - \frac{\kappa}{B} \right) \right]^{1/2}. \quad (5)$$

These are the elliptical eigenpolarizations of the material. Naturally, in the absence of birefringence ($\kappa = 0$) they are purely circular polarization states (i.e., $u = \pm v$).

The solitary-wave solutions to Eqs. (3) correspond to the separatrix trajectories of the potential V , i.e., those trajectories connecting either saddle points or maxima. The separatrices that connect the pairs of opposite saddle points correspond to the linearly polarized NLS dark solitons. Setting $v = 0$ and $u = 0$ in Eqs. (3a) and (3b), respectively, we find that $u = \sqrt{\beta + \kappa} \tanh(\sqrt{\beta + \kappa} x)$ and $v = \sqrt{\beta - \kappa} \tanh(\sqrt{\beta - \kappa} x)$. In isotropic media ($\kappa = 0$) the separatrices that connect opposite maxima lie on the bisecting lines $u = \pm v$ and correspond to the circularly polarized NLS dark solitons. Setting $u = \pm v$ in Eqs. (3), we obtain $u = \pm v = \sqrt{\gamma} \tanh(\sqrt{\gamma} x)$, where $\gamma = \beta/[2(1-B)]$. Note that Eqs. (3) can be normalized with respect to β , and in the following we can set β equal to unity without loss of generality.

In Fig. 1(a) we show a contour plot of $V(u, v)$ in the particular case of an isotropic medium $\kappa = 0$ with $B = 1/3$ (corresponding in the temporal domain to circular silica fibers). The thick solid lines on the axes in Fig. 1(a) connect opposite saddle points, corresponding to linearly polarized dark solitons. The dashed lines that connect opposite maxima represent the circularly polarized dark solitons. From that simple mechanical analog approach it appears natural to address the question of the existence of separatrices that would connect adjacent maxima of the potential

V. We have investigated this problem numerically by means of a standard shooting technique applied to Eqs. (3), which are in general nonintegrable. This technique permitted us to identify trajectories that connect the four adjacent maxima of V . They are represented by the dotted curves in Fig. 1(a). The solitary-wave envelopes $u(x)$ and $v(x)$ corresponding to the separatrix that connects the maxima of the first and fourth quadrants ($u > 0$) are shown in Fig. 1(b) (the envelopes of the separatrices of the other quadrants differ by only changes of the signs of u and v). The envelope u exhibits a small hump on a constant-amplitude background whereas the envelope v switches progressively between two plateaus of identical amplitude but opposite signs. As $x \rightarrow \pm\infty$, both polarization components have the same modulus. In Fig. 1(c) we plotted the corresponding intensity profile $I(x) = u^2 + v^2$ and degree of ellipticity $q = v/u$ (the ratio between the short and the long axes of the polarization ellipse). The values $q = \pm 1$ represent circular polarization states of opposite sense of rotation, whereas $q = 0$ represents a linear polarization. It is apparent that the bound solitary waves (u, v) constitute a localized structure of the field separating two uniform waves of counterrotating circular polarizations. Since the circular polarization states are eigenpolarizations of the isotropic medium considered here, this localized structure can be seen as a domain wall by analogy to the theory of ferromagnetism (the atomic spin being replaced here by the polarization of light). Following the terminology of Ref. 9 (in which similar localized structures are found in systems of counterpropagating beams), we call these new solitary waves polarization domain walls. Let us note that, when B is zero, the potential V has a circular symmetry [see Eq. (4)] and no longer possesses maxima or saddle points. This means that the existence and features of the polarization domain walls directly depend on the tensor character of the Kerr nonlinearity.

As shown in Eqs. (5), the eigenpolarizations of birefringent Kerr materials are no longer simple circular polarization states. Figure 2(a) shows a contour plot

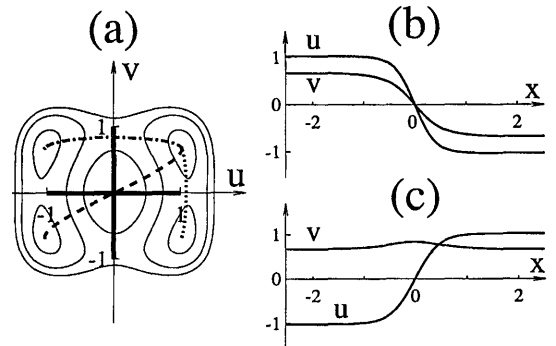


Fig. 2. (a) Contour plot of the potential V for a birefringent medium characterized by $\kappa = 0.2$ ($\beta = 1$). The thick solid lines represent the linearly polarized NLS dark solitons. The dashed curve shows the separatrix of the elliptically polarized dark soliton. The dotted-dashed curve corresponds to the polarization domain wall. (b) The elliptically polarized dark soliton. (c) The polarization domain wall.

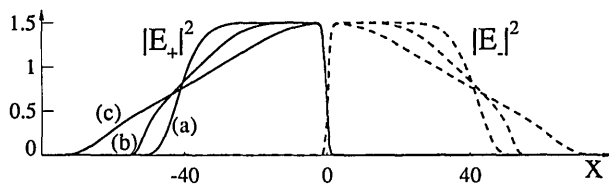


Fig. 3. Numerical simulation of the propagation in an isotropic medium of the polarization domain wall inscribed onto a broader square-shaped pulse. The curves show the intensity profiles of the circular polarization components $E_{\pm}$ at (a) $z = 0$, i.e., initial conditions, (b) $z = 16$, (c) $z = 32$.

of the potential V when $\kappa = 0.2$ (and $B = 1/3$). We see that the birefringence breaks the fourfold symmetry into a twofold symmetry with respect to the u and v axes. As a result, the saddle points are still on the axes, and the corresponding separatrices [thick solid lines in Fig. 2(a)] give the linearly polarized dark NLS solitons mentioned above. The maxima are now closer to the u axis. In fact, from Eqs. (5) we see that the maxima merge with the saddle points of this axis if κ becomes larger than $\beta B/(1-B)$. Note that the normalized intensity $I = u^2 + v^2$ at the maxima is given by $I = \beta/(1-B)$. This shows that the intensity of the waves and the birefringence of the material must be such that $\kappa < BI$ in order to have polarization domain walls in anisotropic media. The trajectories that connect opposite maxima (and that correspond to circularly polarized NLS solitons in the case of isotropic media) no longer lie on the bisecting lines $u = \pm v$ and must be calculated numerically. They are represented by the dashed curve in Fig. 2(a). The envelopes corresponding to this trajectory are given in Fig. 2(b). We see that the solitary wave consists of a transition between two uniform waves of the same amplitude and eigenpolarizations but of opposite signs (i.e., as with the NLS dark soliton there is an abrupt π phase change in both u and v). This solution constitutes a generalization of the circularly polarized NLS dark solitons to birefringent materials.

In the presence of birefringence there are two types of trajectories that connect adjacent maxima. They are represented by the dotted and dotted-dashed curves of Fig. 2(a). The solitary waves correspond-

ing to the dotted-dashed curve are illustrated in Fig. 2(c). They constitute a domain wall between two uniform waves of counterrotating elliptical eigenpolarizations. Naturally, the same situation is obtained with the dotted trajectory (the only difference is that u is now symmetric and v antisymmetric with a smaller amplitude).

Figure 3 shows the numerical simulation of Eqs. (1) with no birefringence ($\kappa = 0$) using a polarization domain wall on a broader pulse as initial conditions. The solid and the dashed curves show the intensity profiles of both circular polarization components $E_{\pm} = (u \mp v)/\sqrt{2}$. The domain wall clearly propagates without distortion, whereas the carrier pulse broadens as a result of diffraction (or dispersion) and nonlinearity.

In conclusion, on the basis of a simple numerical analysis of the coupled NLS equations, we have shown the existence of a novel type of vector dark solitary waves in Kerr materials. These waves constitute localized structures separating regions of different eigenpolarizations of the Kerr medium and can therefore be viewed as polarization domain walls.

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